جامعة خليفة
Khalifa University

RISK MANAGEMENT

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1. Purpose:

The purpose of this procedure is to identify EHS hazards and environmental aspects, assess their risks and impacts, and implement the appropriate control measures for all the activities conducted within KU. This procedure is based on the requirements of ISO 45001:2018 and ISO 14001:2015 and it conforms strictly to Abu Dhabi Occupational Safety and Health System Framework (OSHAD SF).

This procedure aims at preventing, or at a minimum, mitigating accidents and incidents, protecting human health, safety and the environment, and ensuring the necessary resources are allocated in a prioritized manner to ensure that any unacceptable risks identified will be prevented or reduced to acceptable levels

2. Scope:

This procedure is applicable to the following KU operations:

- Academic, Research and Development, Human Resources, Finance and Administration, External Relations, Student Affairs which includes Integrated Systems Management ISO 45001:2018 and ISO 14001:2015.
- President Office: The Office of the President encompasses Executive Leadership, Internal Audit and Strategic Planning & Monitoring Offices including those of the Advisory Panel and Legal. The Office of His Excellency is responsible for ensuring the Vision and Mission of the KU is aligned with the Vision of UAE Leadership and is ultimately accountable to them for the delivery of this vision.
- Academic Department: Engineering, Applied Sciences, Health Sciences, Management, Homeland Security, Research, Graduate Studies and University Library.
- Research and Development: EBTIC, Nanotech, Aerospace and Ankabut.
- Human Resources: Organizational Development, Employee Relations and HR Systems and Services.
- Finance and Administration: Information Technology, Administration, Finance and Facilities.
- External Relations: Advancement, PR and Marketing, Continuing Education, Corporate, government and Relations, Exchange Programs.
- Student Affairs: Registrar, admission, Counseling, Community Relations, Alumni, Student Life and Financial Aid.
- All premises belong to Khalifa University.

3. Responsibilities:

3.1 EHS Steering Committee Representative: Responsible for

- Implement, maintain and update this procedure as required.
- Ensure necessary resources are available for an effective implementation and maintenance of this procedure.
- Reporting to top management on the performance of this procedure and any need for continual improvement.

3.2 Department Directors: Responsible for:

- Ensuring that processes needed for this procedure are established, implemented and maintained.
- Participating in all EHS Risk Assessment Review meetings.

3.3 EHS Manager: Responsible for

- Ensure that this procedure is implemented and maintained
- Reviewing the lessons learned from the implementation of this procedure.
- Ensuring the promotion of awareness of this procedure throughout KU.
- Consolidation of Risk Assessment and carrying out Risk Management activities.
- Overall performance monitoring of this procedure.
- Ensuring that all training requirements are identified through coordination with Department Managers and section managers and fulfilled for an effective implementation.
- Establishing, managing and maintaining all required operational control procedures requirements specified in this procedure within Khalifa University as per Operational Control Procedure KU – FD – EHS – MSM – 12.
- Setting an action plan related to identified health and safety related risks and significant impacts and eliminating / mitigating it to ALARP residual level.

3.4 Laboratory Managers, Laboratory Engineers, Laboratory Instructor, Assistant Laboratory Engineer, Senior Laboratory Technician, Laboratory Technician and Section Coordinators: Responsible for

- Effective implementation of this procedure in their respective Department / Section.
- Identifying, evaluating, and calculating risk levels for all health and safety risks and environmental impacts, and implementation / completion of action plans as per EHS Risk Register within their Department / section.

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- Implementing all required operational control procedures.
- Analyze all EHS incidents and accident statistics prior to the periodic updating of risk register and provide necessary advices to the review section's committees.
- Provide necessary guidance to EHS Inspectors and Section Coordinators for carrying out the risk assessments as proactive measurement.
- Effective implementation and maintenance of this procedure.
- Identify all hazardous activities and perform risk assessment in coordination with EHS Manager, and other concerned personnel.
- Coordinating with other Departments and sections for the implementation and maintenance of this procedure.
- Identifying all hazardous activities in KU and perform related risk assessments.

3.5 EHS Inspector: Responsible for

- Ensure that all necessary control measures identified and recorded in the risk assessment register are implemented and followed at all site operations.
- Monitoring of the effectiveness of this procedure.
- Reporting non-conformities to EHS Management Committee.
- Participating in all reviews and updating of risk assessments.

4. Abbreviation / Definitions:

EHSSC	Environment, Health & Safety Steering Committee
EHSMS	Environment, Health and Safety Management System
MSM	Management System Manual
MPM	Management Procedure Manual
ALARP	As Low As Reasonable Practicable

Hazard: A source or a situation with a potential for harm in terms of human injury or ill-health, damage to property, damage to the environment, or a combination of these.

Hazard Identification: Process of recognizing that a hazard exists and defining its characteristics.

Hierarchy of Control: Is a tool used to eliminate or manage the risks to as low a level as practicable. It is listed below in order of effectiveness.

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Elimination: Removal of hazard. For example, eliminating a requirement to carry out the task, use a piece of equipment or utilize a chemical.

Substitution: Replace the material, plant or work practice with a less hazardous one. e.g. replace a hazardous chemical with a less hazardous one.

Engineering Controls: Engineering the solution to minimize risk is highly desirable as the process reduces the reliance on human behavior to effect long lasting positive change. There are a number of aspects of engineering controls.

- One may redesign the way in which work is performed, modify equipment to change the way a task is performed or engineer change to the process steps to eliminate hazardous activity. One may also completely automate a process where there is minimal or no human interaction.
- One may isolate a hazard by physically guarding the hazard, enclosing the hazard thus preventing human contact, may lock a process/equipment thus preventing access by any unauthorized personnel, and may remove the hazard by engineering means such as ventilation.

Administration Controls: The Administration controls are the procedural aspects of managing hazards, such as planned and preventative maintenance programs, standard operating procedures, lock out/tag out procedures, education and training and the rotation of staff thus minimizing exposure.

Personal Protective Equipment (PPE): Personal Protective Equipment is the last and the least effective control method used. It involves staff wearing appropriate PPE such as steel mesh gloves, safety shoes, aprons, goggles etc., to isolate the person from the hazard. Any breakdown of the system immediately exposes the worker to the hazard. This control method is not highly effective because it relies totally on human behavior. PPE often forms part of the 'short term' control methodology.

Incident: A single event/incident which has caused or could have caused fatality, injury, illness and/or damage (loss) to assets, the environment, reputation or third parties.

Near Miss: A near miss is an event that could have resulted in human injury or damage to property, process or the environment.

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Risk: (In relation to any potential injury or harm) the likelihood and consequence of that injury or harm occurring.

Risk Assessment: Is concerned with the prevention of accidents and ill health and to identify the environmental impacts. It is the key to successful management of Environment health and safety. Generally, regulations are increasingly moving away from a list of "what to do" with the trends in legislation now placing greater emphasis on risk assessments to address the actual risk rather than the perceived risk. This ensures that resources are targeted to the areas where they are most needed and are most effective.

Risk Management: Risk management is: “the culture, processes and structures that are directed towards realizing potential opportunities whilst managing adverse effects”.

Risk Register: A list of hazards, associated risks (pre and post control) and controls, sorted in order of the highest to lowest risk.

Risk Control Plan: A Risk Control Plan is an action plan that sets out how the EHS risks identified in a Risk Assessment will be eliminated or controlled. Risk Control Plans are developed from risk assessments. Risk Control Plans may include a list of:

- Hazards identified
- Risks associated with the hazards identified
- Existing risk controls
- Proposed short-term controls (including proposed completion dates)
- Proposed medium-term controls (including proposed completion dates)
- Proposed long-term controls (including proposed completion dates)
- Person(s) responsible for implementation of controls
- Resources required for implementation of controls
- The proposed review date.

Plant and Equipment Includes:

- Any machinery, equipment, appliance, implement and tool; and

- Any component of any of those things; and
- Anything fitted, connected or related to any of those things.

Chemicals and Substances: Any natural or artificial substance, whether in the form of a solid, liquid, gas or vapor.

Systems of Work: Methods, procedures and practices by which work is to be carried out

Buildings, Structure and Work Environments: Means a place, whether or not in a building or structure, where employees or self-employed persons work, and includes a car, truck, ship, boat, airplane and any other vehicle.

Aspect: The organization has been established and maintains procedures to identify the environmental aspects of its activities, products and services that it can control and over which it can be expected to have an influence, in order to determine those which have or cannot have significant impacts on the environment. The organization ensures that the aspects related to these significant impacts are considered in setting its environmental objectives.

Environment: Surroundings in which an organization operates, including air, water, land, natural resources, flora, fauna, humans, and their interrelation.

Impact: Any change to the environment, whether adverse or beneficial, wholly or partly resulting from an organization's activities, products or services.

Risk Control: That part of risk management which involves the implementation of policies, standards and procedures or physical changes to eliminate or minimize adverse risks.

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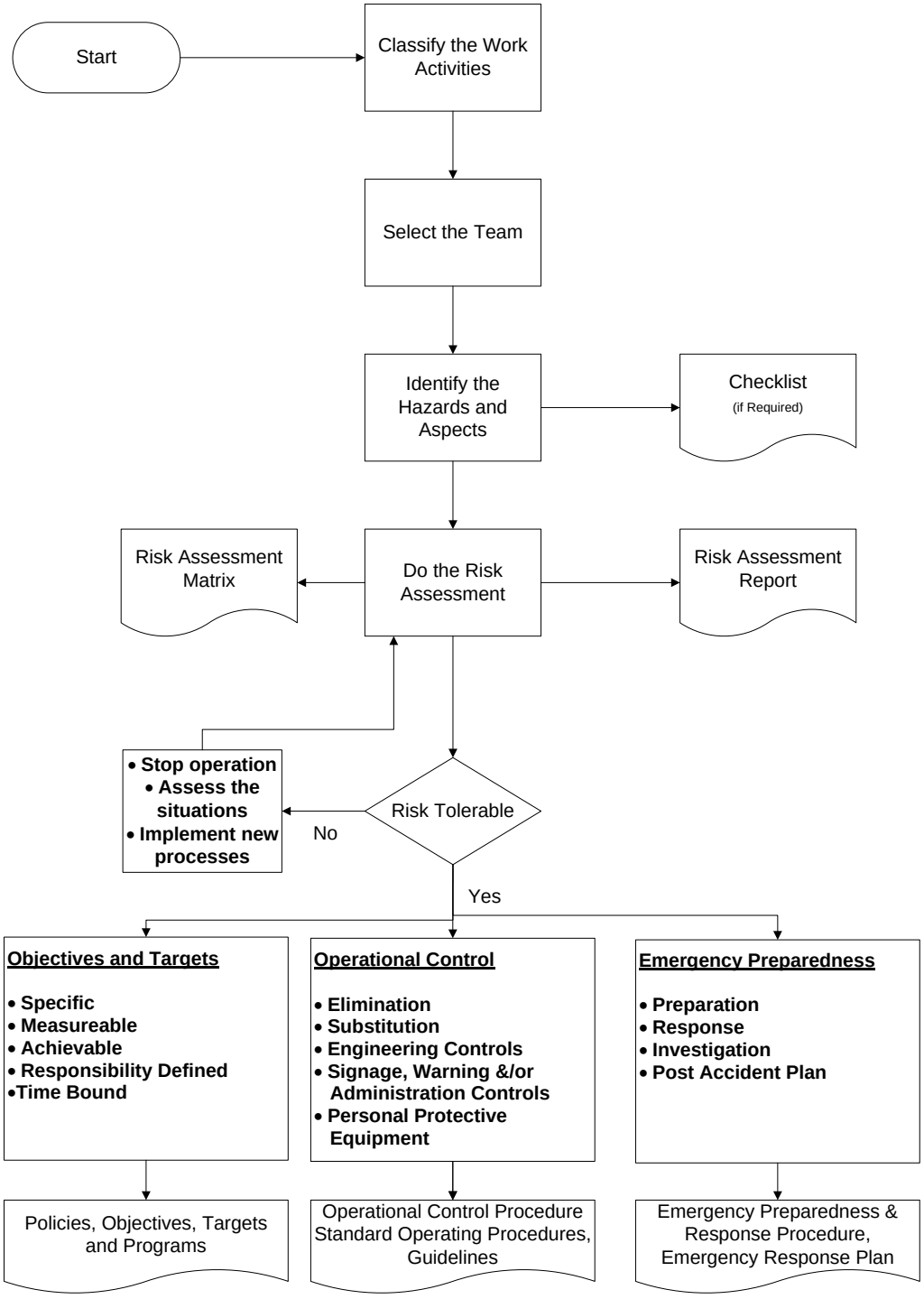
5. Procedure:

5.1 General: the flow chart describes the Risk Management Process:



5.1.1 Identification and Classification of Work Activities:

The very first step for controlling hazards is to identify and classify the work activities and Employees carrying out the activity are usually best placed to identify and classify them into the following categories using form KU – FD – EHS – FOR – 01:

- Routine and non-routine activities ;
- Location of KU's premises, operations and activities;
- Abnormal activities: infrequently conducted operations that are done once a month or less; and
- Emergency situations: including preparedness and response conditions.

5.1.2 Risk Classification of Laboratories:

LOW, No significant risk	Computer/modeling activity; Low voltage (up to 60VDC/40VAC) electrical activity; Low risk rated chemicals – Refer Table 4: Chemical Hazard Categories; BSL 1 Labs– Refer Table 5: Biological Hazard Categories; Class 1,1M, 2 & 2M Lasers– Refer Table 6: Laser Hazard Categories.
MEDIUM	Electrical activity dealing with experiments/projects higher than 60VDC/40VAC up to 350VDC/240VAC; Use of overhead crane; Medium risk rated chemicals – Refer Table 4: Chemical Hazard Categories; BSL2 & BSL3 Labs - Refer Table 5: Biological Hazard Categories; Class 3R, 3B Lasers– Refer Table 6: Laser Hazard Categories.
HIGH	Electrical activity dealing with experiments/projects higher than 350VDC/240VAC; activity that have class IV Laser, Experiments at high pressure (> 300bar) and/or high temperature (> 500 °C); High risk rated chemicals– Refer Table 4: Chemical Hazard Categories; BSL2 & BSL3 Labs - Refer Table 5: Biological Hazard Categories; Class 4 Laser– Refer Table 6: Laser Hazard Categories; Radioactivity and X-ray activities.

Notes:

- Use the higher category, if multiple-classes are applicable.
- Activities with high hazard require task specific risk assessment

5.1.3 Occupational Hazards and Environmental Aspects Identification:

All hazards and aspects for each location/ facility are identified utilizing the “Generic EHS Risk Register”, this process is owned by EHS Section and EHS committee.

All hazards and aspects of each activity are identified utilizing the “Department/ Section EHS Risk Register” This process to be owned by the Departments/ Section and will determine those situations/ activities which could cause harm.

The following factors are considered while performing the initial review:

- Past records and relevant published literature.
- Practice and relevant experience and specialist and expert judgments using economics, engineering or other models.
- Initial EHS gap analysis report.
- Legal and other requirements.
- Any complaints from stakeholders and other interested parties.

5.1.4 The techniques used for identification:

Carrying out Hazard Identification shall be done in a number of ways– such as

- Workplace walk through and observation;
- Structured interviews with experienced employees in the area / activity checked;
- Use of multi-disciplinary groups of experts;
- Individual evaluations using questionnaires;
- Use of models and simulations;
- Employee reporting including accidents and incidents; and
- Reviewing tasks/process step by step.

5.2 Risk & Impact Assessment:

All potential hazards and environmental aspects are identified with the activities, services, and processes and the importance of each hazard or aspect are evaluated with respect to its actual or potential risk or impact.

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5.2.1 Identify Hazards and Aspects and their Consequent Risks and Impacts:

All activities/ areas are visited and studied and all related hazards and aspects (both actual and potential) are identified; then risk assessment has been conducted with related applicable potential incident events.

The study takes into consideration the following areas:

- Land-based ecosystem;
- Aquatic ecosystem;
- Atmosphere/ waste/ other - (air and noise);
- Cultural heritage (indigenous and modern);
- Health effects;
- Human health and safety;
- Production loss; and
- Total cost of impacts or incident event.

The operations that need to be considered include, but are not limited to:

- Work activities, practices and procedures;
- Work equipment and devices;
- Workplace and associated facilities;
- Used materials and substances;
- Work environment related issues, like temperature, lighting, humidity;
- Use of natural resources and energy;
- Emissions to air, releases to water, releases to Land, waste and by products, energy use, energy Emitted (e.g. heat), use of raw materials and natural resources, physical attributes (e.g. size, shape, color, appearance), and University's core activities (e.g. teaching and learning)
- Transportation and disposal of materials and chemicals; and
- Subcontracted activities and operations.

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For all environmental aspects, discharges or emissions of pollutants and other environmentally harmful substances, the impacts are based on factors such as those listed below which help in the assessment of relative consequences:

- Persistence;
- Toxicity;
- Health effects;
- Concentration;
- Volume discharged per event;
- Duration of the discharge;
- Proximity to water-bodies;
- Potential dilution;
- The area of land/marine waters affected; and
- Secondary consequences and existing mitigation measures.

The potential consequences are judged using as much available information as possible.

For the activity screening risk assessment, five levels of severity of consequences have been used – insignificant, minor, moderate, major and catastrophic. The definitions used to assess relative consequences are shown in Table 1.

Table 1: provides a consistent method of assessment that can be applied by different personnel and at different times.

At this stage, the following items have been included in the risk register:

- Functional Area/ Sub-area;
- Activity/ Process;
- Ownership;
- Hazard;
- Aspect; and
- Risks/ Environmental Impacts.

Levels of Risk Assessment:

A staged approach has adopted for risk assessment, involving the following steps:

- Stage 1 or Simple Risk Assessment;
- Stage 2 or Screening Risk Assessment; and
- Stage 3 or Detailed Risk Assessment.

The goal of the staged approach is to use simple conservative assumptions in preliminary assessments to identify which issues are likely to be most relevant with respect to risk, allowing more detailed (site-specific) risk assessment to focus on these issues.

This flexible approach allows resources to be focused on the most critical issues associated with a site or project in a prioritized and defensible manner to ensure that any unacceptable risks will be reduced to an acceptable level.

Estimate the Probability of Occurrence and Severity:

Risks and impacts are analyzed on the basis of probability and severity as follows:

- The probability of occurrence then needs to be estimated. Table 2 provides the probability of occurrence ratings with descriptions for estimating the probability of each occurrence;
- Frequency descriptions have been split into Environment, Health and Safety for clearer explanation;
- The probability of occurrence from any event, mode of occurrence or failure mechanism has been considered;
- In addition to evaluating the frequency for routine and non-routine operations, the following scenarios and conditions are considered:
    - Activities of all persons having access to the workplace (including contractors and visitors);
    - Human behavior, capabilities and other human factors;
    - Identified hazards and aspects originating outside the workplace capable of adversely affecting the health and safety of persons under the control of the organization within the workplace;
    - Hazards and aspects created in the vicinity of the workplace by work-related activities under the control of the organization;

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- Infrastructure, equipment and materials at the workplace, whether provided by the organization or others;
- Changes or proposed changes in the organization, its activities, or materials;
- Modifications to the OH&S management system, including temporary changes, and their impacts on operations, processes, and activities;
- Any applicable legal obligations relating to risk assessment and implementation of necessary control; and
- The design of work areas, processes, installations, machinery/equipment, operating procedures and work organization, including their adaptation to human capabilities.

At this stage, the existing controls related to the operation are recorded in the EHS Risk Register, including:

- Legal/ regulatory;
- Guidelines, engineering and Safety Controls;
- EHS Policy / Objectives;
- Availability of Resources;
- Costs and Benefits; and
- Technical Knowledge.

5.2.2 Risk Evaluation:

Risk levels are being calculated by multiplying the Consequence Score and Probability of Occurrence together.

Risk = Consequence Score X Probability of Occurrence

Table 3: shows the risk matrix and the established risk categories.

The final outcome is in relative point scores, rather than actual risks.

Note that risk estimations have been carried out assuming a well-managed installation operating under normal conditions. Risk scores do not reflect the higher risks under adverse conditions/ worst-case scenarios.

The relative risk score is estimated to arrive at a definition of those risks that are significant, and those that are of a lesser nature. This allows a better understanding of the risks and their evaluation.

Four risk categories have been used**: Extreme, High, Moderate,** and **Low**. Table 3 shows the risk matrix and the established risk categories.

At this stage, the existing controls related to the operation are being recorded in the EHS Risk Register, including:

- Probability;
- Consequence; and
- Risk / Impact Rating.

The assessment is based on the Risk Assessment Matrix – Table 3

5.2.3 Risk Assessment Reporting Training:

After completing the EHS Risk Register, the following is completed:

- Additional Risk/ Impact Controls;
- Recommendations; and
- Responsibility.

All necessary operational controls are referenced within the EHS Risk Register in the recommendations section, and their implementation is being overseen by the EHS Section to bring all risks to acceptable levels. For all nonconformities, necessary actions shall be taken after carrying out the risk assessment.

5.2.4 Preparing and Implementing Operational Control Measures:

Preventive/ Corrective actions are identified in order to manage the hazard/ aspect at an acceptable and “as low as reasonably practicable” (ALARP) risk level. For hazards with low risk, actions may not be required.

The health and safety hierarchy of control is as follows:

- Elimination of the hazard (where possible);
- Substitution (e.g. Use a less hazardous substance/ environmentally friendly materials);
- Engineering / isolation - for plant and equipment (e.g. Exhaust ventilation);
- Procedural (e.g. Via improving work methods, housekeeping); and
- Personal protective equipment (PPE) – it is the last resort if the above measures are not practicable.

The environmental hierarchy of control is as follows:

- Avoidance/ Source Reduction;
- Reuse;
- Recycling;
- Recovery of Energy;
- Treatment;
- Containment;
- Disposal; and
- Remediation and cleanup.

Any required operational control procedures are established to prevent or mitigate the identified risks and impacts. These operational control procedures contain the emergency response requirements of such risks and impacts. When the prevention/ mitigation measures are identified, control measures have been established, addressing roles and responsibilities, training required for the relevant parties, time frame for completing the actions, the required changes for the EHSMS and its associated documents/ procedures, the procedures for quality assurance, monitoring, maintenance and inspection as applicable.

In determining the acceptable level of risk, the level of control must consider the cost involved, the time to implement and overall benefit. For example, expending the high cost to control a hazard that has a very low likelihood of occurring and low severity of outcome would not be considered acceptable and Khalifa University EHS Steering committee would make an informed decision on the basis of the risk assessment to accept the risk. However, under no circumstances shall cost alone be used as a reason to compromise safety.

Implementation Working Group includes a representative from each concerned Department will be established in order to follow up the implementation of corrective actions.

An electronic database has been established and access granted for each representative. Department's representatives will update the information.

5.2.5 Monitoring and Review:

Ongoing review of the Risk Management Procedure is being conducted to ensure that the EHS performance of KU and the developed EHS implementation plans

remain relevant. Factors that may affect the probability and consequences of an outcome may change, as the factors that affect the suitability or the cost of the mitigation options. It is therefore necessary to repeat the risk management cycle regularly. Actual progress against EHS Risk Register provides an important performance measure and is incorporated into the KU performance management, measurement and reporting system. Monitoring and review also involve learning lessons from the risk management process, by reviewing events, the mitigation plans and their outcomes.

As per EHSMS implementation plan register, refer doc. No.: KU – FD – EHS – REG - 03, all related activities are implemented, monitored and reviewed periodically. EHS Risk register, doc. No.: **KU – FD – EHS – REG – 01** is reviewed, updated twice a year and continuously monitored. Department wise progress for eliminating / mitigating risks as per EHS Risk register is continuously monitored.

5.3 Risk Assessment Methodology:

Risk analysis has been carried out using a combination of Quantitative and semi-quantitative assessment methodology:

Table 1: Potential EHS Risks and Impacts (with Potential Incident Consequences) Rating

Area impacted (a)	Insignificant consequences (Score = 1)	Minor consequences (Score = 2)	Moderate consequences (Score = 3)	Major consequences (Score = 4)	Catastrophic consequences (Score = 5)
Environment					
Land-based Ecosystem	Insignificant environmental impact. Occasional damage by erosion, or of flora and fauna. Some disruption to flora or fauna habitats.	Minor impacts on fauna/flora and habitat, but no negative impacts on ecosystem function. Limited damage to a minimal area of land of no significant value (i.e. No unique habitats). Temporary damage/ disruption (<1 month) to flora or fauna habitats.	Significant changes in flora/ fauna populations and habitat. Disruption to, or some death of, rare flora or fauna, but not resulting in eradication of endangered species. Non-persistent but possibly widespread damage to land: damage that can be remedied without long-term loss; localized persistent damage; or significant temporary damage (<1 year) to the ecosystem.	Continuous and serious damage by erosion or to flora or fauna. Major disruption to, or frequent death of, rare flora or fauna. Major destruction of significant habitat.	Long-term and significant change in population (e.g. Eradication of endangered species) or habitat with negative impact on ecosystem function. Widespread destruction to a significant area of land, rare flora and fauna and/or groundwater resource.

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Area impacted (a)	Insignificant consequences (Score = 1)	Minor consequences (Score = 2)	Moderate consequences (Score = 3)	Major consequences (Score = 4)	Catastrophic consequences (Score = 5)
Aquatic Ecosystem	Occasional short-term impact and/or disruption to aquatic flora and fauna.	Minor impact on aquatic ecosystem, including flora, fauna and habitat. No significant impact on water resources.	Significant localized impacts but without the long-term impact on aquatic ecosystems, and/or short term impacts on water resources.	Significant widespread impact on protecting wildlife (e.g. Migratory shorebirds, marine plants, fish breeding grounds), or aquatic ecosystems of moderate duration.	Damage to an extensive portion of aquatic ecosystem resulting in severe impacts on aquatic populations and habitats and/or long-term impact on water resources.
Atmosphere/ Waste/Other (Air and Noise)	Temporary nuisance from noise, odor, dust, other air emissions, greenhouse gases, vibration, visual impact. Minor use of water, fuels and energy and other natural resources.	Minor environmental impact due to contained release of pollutant (including odor, dust and noise), fire or explosion with no lasting detrimental effects. No outside assistance required.	Creation of noise, odor, dust, other controlled/ uncontrolled air emissions, greenhouse gases, vibration, and visual impact at significant nuisance levels.	Major environmental impact due to uncontained release, fire or explosion with detrimental effects. Outside assistance required.	Catastrophic environmental impact due to uncontained release, fire or explosion with detrimental effects. Outside assistance required. Extensive chronic discharge of persistent hazardous pollutant.
Waste	Results in the generation of significant quantities of non-hazardous wastes.	Significant use of water, fuels and energy and other natural resources.	Results in the generation of significant quantities of hazardous wastes.	Major environmental impact due to uncontained release, fire or explosion with detrimental effects. Outside assistance required.	Results in the generation of significant quantities of intractable wastes.
Cultural heritage (indigenous and modern)	Minor repairable damage to commonplace structures.	Minor repairable damage to structures/ items of cultural significance, or minor infringements of cultural values.	Moderate damage to the structures/ items of cultural significance, or significant infringement of cultural values/ sacred locations.	Major damage to structures/ items of cultural significance, or major infringement of cultural values/ sacred locations.	Irreparable damage to highly valued structures/ items/ locations of cultural significance or sacred value.
Health					
Health Effects	Insignificant impact on employees, and/or surrounding communities.	Minor complaints or exposure during non-routine activities like breakdown maintenance. Maximum occurrence limited to two times per year	Ongoing complaints from community. Significant emission or discharge that impacts on surrounding populations.	Major ongoing long-term health effects likely to surrounding communities and workers.	Extreme health risk – potential for death in community.
Safety					
Human Health and Safety	Near miss incident or trivial injuries, which may require self-administered first aid. Injured personnel can continue to perform normal duties.	Injuries requiring on-site treatment by a medical practitioner. Personnel unable to continue to perform duties.	Serious injuries requiring off-site treatment by a medical practitioner or immediate evacuation to hospital. Potential long-term or permanently disabling effects.	Single fatality.	Multiple fatalities.

Area impacted (a)	Insignificant consequences (Score = 1)	Minor consequences (Score = 2)	Moderate consequences (Score = 3)	Major consequences (Score = 4)	Catastrophic consequences (Score = 5)
Production Loss	Incident event without causing production loss.	Production loss or delay up to one week.	Production loss or delay of one week to one month.	Production loss or delay for over one month.	Loss of license to operate or ability to produce indefinitely.
Total Cost of Impacts or Incident Event	Financial loss (compensation, fines, cost to repair, plant damage) of less than AED5, 000.	Financial loss (compensation, fines, cost to repair, plant damage) of AED5, 000 - AED50, 000.	Financial loss (compensation, fines, cost to repair, plant damage) of AED50,000 – AED100,000	Financial loss (compensation, fines, cost to repair, plant damage) of AED100,000 - AED 500,000	Severe financial penalties or legal liabilities. Financial loss (compensation, fines, cost to repair, plant damage) of greater than AED 500,000.

Table 2: Probability of Occurrence

Descriptor	Likely Frequency		Probability
	Environment	Health and Safety	
Frequent	Continuous or will happen frequently.	Occurs several times a year at locations	5
Often	5 – 12 times per year	Occurs several times a year in similar industries in UAE	4
Likely	1 – 5 times per year	Has occurred at least once in UAE	3
Possible	Once every 5 years	Has occurred in industry (Worldwide)	2
Rare	Less than once every five years	Never encountered in industry	1

Table 3: Risk Categories

Probability (From Table 2.)	Consequence (From Table 1.)				
	Insignificant (1)	Minor (2)	Moderate (3)	Major (4)	Catastrophic (5)
Rare (1)	1	2	3	4	5
Possible (2)	2	4	6	8	10
Likely (3)	3	6	9	12	15
Often (4)	4	8	12	16	20
Frequent/Almost certain (5)	5	10	15	20	25

Risk Categories Definition		
15 - 25	**Extreme Risk**	Activity or industry should not proceed in current form
8 - 12	**High risk**	Activity or industry should be modified to include remedial planning and action and be subject to detailed EHS assessment
4 -6	**Moderate risk**	Activity or industry can operate subject to management and /or modification
1 - 3	**Low risk**	No action required, unless escalation of risk is possible

Table 4: Chemical Hazard Categories

Hazard Category	Definition	Risk Level LOW	Risk Level MODERATE	Risk Level HIGH
Acutely Toxic Liquids and Solids	Any substance that can cause minor or serious illness or death on exposure. These substances have a GHS classification of acute toxicity Cat 1-3 and are identified by the skull and crossbones pictogram. *Due to the varying quantities of chemicals and their toxic effects, we do not define the risk level by quantities of acutely toxic chemicals.	N/A	Presence of Category 3 acute toxins.	Presence of Category 1 or 2 acute toxins.
Compressed Gases – Inert	Inert gases are non-toxic and non-flammable but pose hazards because they can displace oxygen, are highly pressurized, and can be extremely cold or cryogenic.	√	N/A	N/A
Compressed Gases – Flammable	A flammable gas is any material that is a gas at 20°C (68°F) or lower and is also ignitable when in a mixture of 13% or less by volume with air, or that has a flammable range with air of at least 12% regardless of lower limit.	≤5%	5< 75% composition	75% ≤ composition
Compressed Gases – Oxidizing	Oxidizing gas initiates or promotes combustion of materials, either by catching fire itself or by causing a fire through the release of oxygen or other gases.	√	N/A	N/A
Compressed Gases – Toxic	Toxic gases are poisonous or can cause death.	N/A	N/A	√

Hazard Category	Definition	Risk Level LOW	Risk Level MODERATE	Risk Level HIGH
Corrosives	Corrosive materials are chemicals that cause visible destruction or irreversible alterations in living tissue by chemical action at the site of contact.	Presence of < 10 L dilute (<1M) corrosive liquid. < 5 L concentrated weak corrosive liquid. < 100 g of corrosive solids.	Presence of 10-20 L dilute (<1M) corrosive liquid. 5-10 L concentrated weak corrosive liquids. < 5 L concentrated strong corrosive liquids. 100 g -1 kg corrosive solids.	Presence of > 20 L dilute (<1M) corrosive liquid. > 10 L concentrated weak corrosive liquid. ≥ 5 L strong corrosive liquids. > 1 kg of corrosive solid.
Cryogens	Materials at ultra-low temperatures (< -73.3 °C) are considered cryogenic. Hazards include freezing of tissue, asphyxiation, toxicity, high pressure gas hazards, flammability, and explosion hazards.	Presence of small quantities (<5 L) of cryogens in well-ventilated areas. Examples: 1 gallon bucket use of dry ice.	Presence of moderate quantities (5-20 L) of cryogens in well ventilated areas. Examples: Use of a 10 L Dewar of liquid nitrogen.	Presence of cryogens in excess of 20 L. Presence of cryogens that are toxic, highly flammable, or oxidizers. Examples: Use of liquid oxygen.
Explosive Materials	Explosive substances are solids or liquids (or a mixture of substances) that are capable by a chemical reaction of producing gas at a temperature and pressure and at a speed to cause damage to the surroundings. Compounds classified as explosive, self-reactive, or organic peroxides fall under this category. They are labelled either with the explosive symbol or the flame symbol.	Chemicals that can be explosive in their pure form but that are shipped and stored in a stabilized form (e.g. wetted) and that are NOT labelled with the explosive symbol on the container.	Presence of less than 100 g of material labelled with the explosive symbol.	Presence of 100 g or more of material labeled with the explosive symbol.

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Hazard Category	Definition	Risk Level LOW	Risk Level MODERATE	Risk Level HIGH
Flammable Liquids	Flammable liquids have a flashpoint equal to or less than 200 °F. Most solvents are flammable and can cause irritation and defatting when in contact with skin. Inhalation of vapours can cause dizziness and a headache. Note: the low, moderate, and high storage quantities are based on a conservative interpretation of NFPA 45. If you use large quantities of flammable liquids, contact DRS for clarification.	Presence of flammable liquids less than 2 L/ 100 ft2.	Presence of flammable liquids in the range of 2-6 L/100 ft2.	Presence of flammable liquids > 6 L/100 ft2.
Flammable Solids	Flammable solids are solids that are readily combustible or that may cause or contribute to fire through friction. They can be easily ignited by brief contact with an ignition source, and the flame spreads rapidly. Pyrophorics and water-reactive solids will be captured in separate categories. category 1 materials are rapidly burning metals and compounds very resistant to being extinguished by water; category 2 materials are slower burning metals and compounds that are susceptible to extinguishment by water.	Category 2 flammable solids	Presence of less than 500 g of Category 1 flammable solid	Presence of ≥500 g Category 1 flammable solids.
Health Hazards	Compounds that are identified as mutagens, carcinogens, reproductive toxins, target organ toxins, or respiratory sensitizers. Under GHS, these compounds are identified by pictogram 1807 (“the exploding man”). Please note that this pictogram also describes other health hazards that do not fall under this category (e.g., aspiration hazard).	N/A	Compounds classified as carcinogen, mutagen, or reproductive toxin category 2. These compounds are suspected to cause carcinogenic tumors, genetic mutation, or reproductive defects	Compounds classified as carcinogen, mutagen, or reproductive toxin category 1a or 1b. These compounds are presumed or known to cause carcinogenic tumors, genetic mutation, or reproductive defects

Hazard Category	Definition	Risk Level LOW	Risk Level MODERATE	Risk Level HIGH
Oxidizers – Liquids and Solids	A chemical not necessarily combustible that can initiate or promote combustion in other materials, thereby causing fire either of itself or through the release of oxygen or other gases.	Presence of < 100 g	Presence of 100 – 2000 g	Presence of > 2000 g
Peroxide forming chemicals	Chemicals susceptible to formation of explosive peroxides over time when exposed to light or oxygen. Depending on the explosion hazard of the peroxides formed, peroxide-forming chemicals are divided into 3 groups.	Presence of < 100 g listed under Category C. Storage of < 1000 g listed in Category B.	Presence of 100 g – 1000 g listed in Category C. Storage of ≥ 1000 g listed in Category B.	Presence of ≥1000 g listed under Category C. Storage of any amount listed under Category A
Pyrophorics	A gas, liquid, or solid that ignites spontaneously in air.	N/A	N/A	Presence of any quantity.
Water Reactive	Materials that react, sometimes violently, on contact with water, releasing toxic or flammable gas.	Presence of < 100 g	Presence of 100 – 1000 g	Presence of > 1000 g
Nanomaterials	Engineered materials on the nano-scale with any external dimension less than 100 nm that possess properties different than the bulk material. Biological materials such as proteins and DNA do not fall under this category.	Non-destructive use of nanomaterials that are bound in a solid matrix or gel or permanently bonded to a substrate with no potential for release. Example: using a glass slide with metal nano particles on the surface for optoelectronic measurements.	Use of nanomaterials suspended in solvents with potential for release as aerosols. Example: Sonication of an aqueous suspension of quantum dots.	Use of dry nanomaterials in the state of powders and pellets with high potential for release into air. Example: Weighing or transferring dry carbon nano tubes.

Table 5: Biological Hazard Categories

Hazard Category	Definition	Risk Level LOW	Risk Level MODERATE	Risk Level HIGH
BSL1	Agents that present minimal potential hazard to personnel and the environment.	√	N/A	N/A
BSL2	Agents associated with human disease & pose moderate hazards to personnel and the environment.	N/A	√	N/A
BSL3	Indigenous or exotic agents, agents that present potential for aerosol transmission, & agents causing serious or potentially lethal disease.	N/A	√	N/A
BSL4	Dangerous & exotic agents that pose a high risk of aerosol transmitted laboratory infections and life threatening disease.	N/A	N/A	√

Table 6: Laser Hazard Categories

Hazard Category	Definition	Risk Level LOW	Risk Level MODERATE	Risk Level HIGH
Class 1	Low energy levels, not hazardous to skin or eyes. Safe during normal operation.	√	N/A	N/A
Class 1M	Safe during normal operation, but may cause eye injury if viewed with an optical instrument.	√	N/A	N/A
Class 2	Visible wavelengths only, blink response provides eye safely. Will not burn skin.	√	N/A	N/A
Class 2M	Visible wavelengths only, blink response provides eye safely for unaided viewing but can be hazardous if view through optical instruments. Will not burn skin.	√	N/A	N/A
Class 3R	Transitional zone between safe and hazardous laser products. Direct viewing of beam may be hazardous as well as certain specular reflections.	N/A	√	N/A
Class 3B	Direct viewing and specular reflections can cause eye injury. Diffuse reflections are usually safe.	N/A	√	N/A
Class 4	Can cause severe skin and eye injury through and direct exposure, specular reflections and sometimes from diffuse reflections. Often a fire hazard as well	N/A	N/A	√

6. Reference Standards:

6.1 ISO 14001:2015 Environmental Management System – Requirements. 6.1.2.

6.2 ISO 45001:2018 – Occupational Health and Safety Management Systems – Requirements. 4.3.1.

6.3 Abu Dhabi Occupational Safety and Health System Framework (OSHAD SF) – Element 02 – Risk Management.

7. Annexure:

7.1 **Annexure 1** – Risk Assessment Form doc. No.: KU– FD – EHS – FOR – 02.

8. Records / Documents / Formats:

8.1 EHS Risk Register, Dec. No.: KU – FD – EHS – REG – 01.

8.2 EHS Risk Register Format No.: KU – FD – EHS – FOR – 03.

8.3 Car Parking Risk Assessment Checklist: KU – FD – EHS – CHE – 01.

8.4 Fire and Emergency Risk Assessment Checklist: KU – FD – EHS – CHE – 02.

8.5 Building General Risk Assessment Checklist: KU– FD – EHS – CHE – 03.

8.6 Identification and Classification of Work Activities: KU– FD – EHS – FOR – 01.

8.7 Risk Assessment Form doc. No.: KU– FD – EHS – FOR – 02.

8.8 Class Room Checklist: KU – FD – EHS – CHE – 04.

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Record of Reviews & Revisions

REV.	DATE ISSUED	DESCRIPTION OF REVISION		REVIEWED BY	AUTHORIZED BY
		PAGE NO	COMMENT		
01	07/06/2018	All	Change of logo	Roberto M.	Belal I.
		Throughout	Change from KUSTAR to KU	Roberto M.	Belal I.
		Throughout	Change the reference from AD EHSMS to OSHAD SF	Roberto M.	Belal I.
		02	Prepared by Reviewed / Updated by, Endorsed for Approval by, and Approval & Issue by	Roberto M.	Belal I.
		21	Change the reference from AD EHS Center to OSHAD	Roberto M.	Belal I.
		21	Reference Standard from ISO 14001:2004 to ISO 14001:2015	Roberto M.	Belal I.
		21	Reference Standard from OHSAS 18001 2007 to ISO 45001:2018	Roberto M.	Belal I.
		21	OSHAD SF Element changed	Roberto M.	Belal I.
02	25/02/2019	11	Risk Classification of Laboratories	Ahammed Basheer PKM	Belal I.
		22	Table 4: Chemical Hazard Categories	Ahammed Basheer PKM	Belal I.
		26	Table 5: Biological Hazard Categories	Ahammed Basheer PKM	Belal I.
		26	Table 6: Laser Hazard Categories	Ahammed Basheer PKM	Belal I.

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